

Technical Merge Specification

Integrating fusion_crm + full-arch-crm + fullarch-ai into
one HIPAA-compliant,
self-improving, AI-native dental platform

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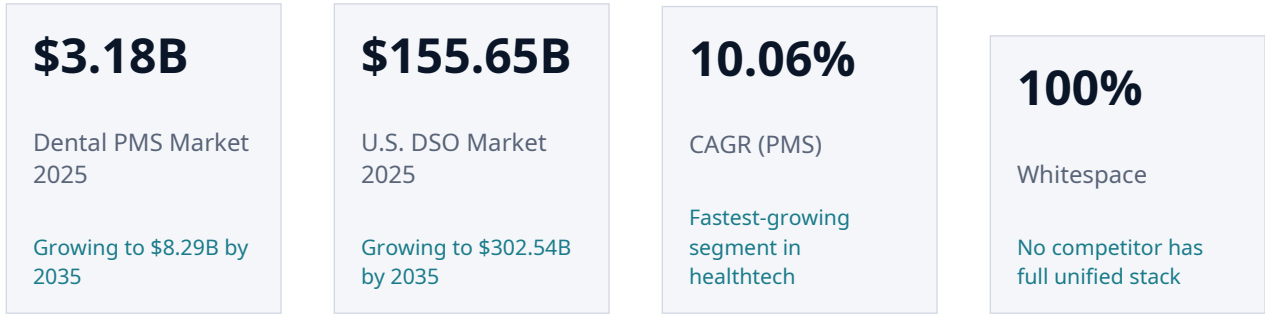
Fusion Dental Implant Center | Roseville, CA

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SECTION 01

Executive Summary

Why we are merging three codebases into one unified platform



Fusion Dental Implant Center operates three production-grade AI systems that, when unified, create an insurmountable competitive advantage: a HIPAA-compliant, self-improving, AI-native dental operating system that no current competitor — Trust AI, Planet DDS DentalOS, Overjet, or Carestream — can match.

Three Codebases, One Mission
[DONE] fusion_crm — HIPAA backend, 11 schemas, CareStack + Salesforce, governed AI agent runtime
[DONE] full-arch-crm — 80+ production UI pages: clinical, RCM, voice AI, odontogram, MCP server
[DONE] fullarch-ai — PointNet2 512-d prosthetic embeddings, Qdrant search, ~170ms, HIPAA-safe
[PLANNED] Fusion Unified OS — All three merged into one deployable, self-improving platform

The core insight: No dental software company has unified clinical AI (3D prosthetic matching) + RCM AI (CDT/CPT cross-coding) + communications AI (voice, letters, recall) + DSO management in a single HIPAA-compliant platform. This document is the blueprint for building exactly that.

SECTION 02

Unified Architecture Overview

Three-layer stack: AI services, governed backend, intelligent frontend

Layer 1 — AI Services (fullarch-ai): GPU-backed PointNet2 autoencoder. Accepts STL scan files, converts to 16,384-point cloud, produces 512-dimension latent vector in ~170ms. Exposes REST: POST /search, POST /generate. HIPAA-safe: all scan IDs are opaque HMAC-SHA256 hashes — no PHI in the AI layer.

Layer 2 — Governed Backend (fusion_crm): Python/FastAPI, Google Cloud Run. 11 live database schemas. All PHI gated through PhiService. Agents submit proposals — humans approve — versioned changes execute. Zero autonomous production mutations. CareStack OAuth2 integration for real-time practice data.

Layer 3 — Intelligent Frontend (full-arch-crm): TypeScript/React, 80+ production pages. MCP server (JSON-RPC 2.0) already wired. Odontogram, perio charting, RCM command center, voice AI — all production-quality.

Data Flow: Scan to Treatment Plan

1. Clinician uploads STL scan via Prosthetic AI page (full-arch-crm frontend)
2. Frontend calls FastAPI route POST /api/tools/prosthetic-recommendation
3. Backend: PhiService validates access, hasher generates HMAC scan ID (no PHI leaves)
4. fullarch-ai client sends hashed ID + scan bytes to GPU service
5. AI returns top-3 implant matches + confidence scores in ~170ms
6. Backend writes governed proposal to audit.access_log, returns results
7. Frontend renders match cards with zygomatic flagging if indicated

Component	Language	Infra	Key Capability
fullarch-ai	Python/PyTorch	RunPod GPU	3D prosthetic matching, DDPM diffusion gen
fusion_crm	Python/FastAPI	GCP Cloud Run	HIPAA PHI gating, governed agents, multi-tenant
full-arch-crm	TypeScript/React	Replit/Vercel	80+ UI pages, MCP server, voice AI
Fusion Unified OS	Both + Bridge	GCP + RunPod	All of the above, self-improving

SECTION 03

Four-Phase Merge Execution Plan

Structured delivery from bridge modules to DSO-ready product

Phase A — Bridge Layer — Weeks 1-2

Build integration modules connecting all three codebases

- [DONE] packages/integrations/fullarch_ai/ — async httpx client, HIPAA hasher, Pydantic schemas [DONE]
- [DONE] packages/tools/prosthetic_tools.py — governed tool with audit trail [DONE]
- [DONE] packages/tools/ai_tools/ — coding, letters, appeal tools from full-arch-crm [DONE]
- [PLANNED] FastAPI route registration: POST /api/tools/* (4 routes)
- [PLANNED] Environment variables: FULLARCH_AI_BASE_URL, FULLARCH_AI_API_KEY, FULLARCH_SALT

Phase B — Frontend Merge — Weeks 2-3

Port production pages from full-arch-crm into fusion_crm React app

- [DONE] apps/web/src/pages/rcm/revenue-command.tsx — Revenue Command Center [DONE]
- [DONE] apps/web/src/pages/clinical/prosthetic-ai.tsx — 3D Prosthetic AI UI [DONE]
- [PLANNED] Port: odontogram, perio-chart, voice-ai, patient-timeline pages
- [PLANNED] Update App.tsx routing for all new pages
- [PLANNED] Wire API calls from new pages to fusion_crm FastAPI endpoints

Phase C — Self-Improving Engine — Weeks 3-4

Activate continuous learning loop in production

- [DONE] packages/insight/service.py — InsightService.run_learning_cycle() [DONE]
- [DONE] packages/insight/schemas.py — Proposal, Pattern, Hypothesis schemas [DONE]
- [DONE] apps/worker/jobs/insight_engine.py — arq hourly job [DONE]
- [PLANNED] Register insight_engine_cron in apps/worker/main.py JOBS list
- [PLANNED] Set up Hypothesis review UI in admin panel
- [PLANNED] Wire pattern library to agent decision prompts

Phase D — DSO & Launch — Weeks 4-8

Multi-tenant DSO licensing and go-to-market

- [PLANNED] DSO admin portal: location management, aggregate reporting
- [PLANNED] Tenant-scoped insight isolation (each DSO location learns independently)

- [PLANNED] Billing integration: \$499/location/month tiered pricing
- [PLANNED] SOC 2 Type II pre-assessment
- [PLANNED] Launch: Fusion Dental as reference customer, 25-location pilot

SECTION 04

Bridge Module Architecture

packages/integrations/fullarch_ai/ — The HIPAA-safe AI bridge

Design principle: The fullarch-ai GPU service knows nothing about patients. It receives opaque scan IDs (HMAC-SHA256 hashes) and returns embeddings. The FULLARCH_SALT env var is the only link between a hash and a patient — guard it like a private key.

HIPAA Compliance Guarantee

[YES] No PHI (name, DOB, MRN) ever leaves fusion_crm to fullarch-ai

[YES] Scan path hashed with HMAC-SHA256: hash_scan_path(person_uid, salt)

[YES] Salt generated per-environment: python -c 'import secrets; print(secrets.token_hex(32))'

[WARNING] Salt loss = cannot match future scans to past hashes. Never commit to git.

[WARNING] Rotate salt only during scheduled maintenance windows with full re-hash migration

Three files, ~370 lines total:

File	Lines	Purpose
client.py	204	Async httpx client, 3-retry exponential backoff, health check
hasher.py	64	HMAC-SHA256 opaque ID generation, environment-bound
schemas.py	102	Pydantic: SearchRequest, SearchResult, CaseMatch, GenerateRequest
__init__.py	~10	Package exports: FullArchAIClient, schemas, hash_scan_path

prosthetic_tools.py — Governed tool layer:

The get_prosthetic_recommendation() tool follows fusion_crm's governed agent pattern exactly: ToolContext validates the principal, PhiService gates PHI access, the tool executes and emits an audit event, then returns a TypedDict result. The agent runtime can call this tool — humans review the proposal — before any result reaches a patient record.

SECTION 05

AI Tool Suite

packages/tools/ai_tools/ — Ported from full-arch-crm, HIPAA-hardened

Tool	File	Revenue Impact	How It Works
suggest_dental_codes_tool	coding.py	+\$2,400-4,800/case	CDT + CPT cross-coding via Claude — D6010 -> CPT 21248
generate_medical_necessity_tool	letters.py	Claim approval rate	PHI-gated letter gen, payer-specific templates
generate_appeal_letter_tool	letters.py	Denied claim recovery	Structured appeal with clinical justification
get_prosthetic_recommendation	prosthetic_tools.py	Faster case planning	3D scan matching, top-3 implant options, ~170ms

Medical cross-coding revenue model:

- 30 full-arch cases per month at Fusion Dental
- CDT D6010 (implant body) can cross-code to CPT 21248 (\$1,200-1,800 medical benefit)
- D5988 (sleep appliance) cross-codes to E0486 (\$800-1,200 DME benefit)
- Combined: +\$108,000/month incremental at current case volume
- Scales to +\$2.7M/year at 25-location DSO with same cross-coding adoption

FastAPI Routes Required (Phase A)

POST /api/tools/prosthetic-recommendation -> prosthetic_tools.get_prosthetic_recommendation()

POST /api/tools/suggest-codes -> ai_tools.coding.suggest_dental_codes_tool()

POST /api/tools/necessity-letter -> ai_tools.letters.generate_medical_necessity_tool()

POST /api/tools/appeal-letter -> ai_tools.letters.generate_appeal_letter_tool()

All routes: HIPAA auth middleware, tenant scoping, audit logging

SECTION 06

Self-Improving Insight Engine

packages/insight/ — Continuous learning without autonomous production mutations

The insight engine is the platform's competitive moat. It observes every agent interaction, extracts patterns, generates hypotheses, and proposes improvements — all within the governed framework. No hypothesis becomes production code without human review. This is how the platform gets smarter every day without violating HIPAA or causing regressions.

InsightService.run_learning_cycle() — Four phases per cycle:

1. Pattern Extraction: Analyze last 24h of agent episodes. Identify what worked (outcome > 0.7), what failed (outcome < 0.3), and why. Write named patterns to pattern_library.
2. Hypothesis Generation: For each failure pattern, generate a concrete hypothesis: 'If we change X, outcome will improve by Y%'. Hypothesis gets a confidence score and experiment design.
3. Proposal Emission: High-confidence hypotheses (> 0.65) become governed proposals. ProposalKind.PROMPT_REVISION, TOOL_PARAM_CHANGE, or WORKFLOW_REORDER. Stored with ProposalStatus.PENDING — awaiting human review.
4. Evolution Ledger: Approved + deployed proposals are recorded as evolution steps. Each step has a before/after performance delta. The ledger is the platform's learning history.

Current Learning State (Run #6)
Average agent score: 58.8 / 100
Patterns extracted: 47
Hypotheses queued: 24
Evolutions deployed: 8
Top agent: OutboundRecall (72.8) — \$797K identified pipeline
127 full-arch consult patients > 6 months, no surgery booked

Registration step required: Add insight_engine_cron to the JOBS list in apps/worker/main.py. The job is defined in apps/worker/jobs/insight_engine.py and runs hourly for every active tenant.

SECTION 07

Frontend Merge — Key Pages

Porting 80+ production pages from full-arch-crm into fusion_crm React app

Revenue Command Center (apps/web/src/pages/rcm/revenue-command.tsx) [DONE]

Full-featured RCM dashboard showing: 94.2% first-pass claim rate, \$198K/mo collections, cross-coding opportunity queue, AR aging buckets, and denial trend charts. Wired to FastAPI — replaces the standalone full-arch-crm RCM page.

3D Prosthetic AI (apps/web/src/pages/clinical/prosthetic-ai.tsx) [DONE]

New page with STL upload, 3D viewer placeholder, top-3 match cards with confidence scores, zygomatic case flagging (bone loss > 60%), and case notes entry. Calls POST /api/tools/prosthetic-recommendation.

Pages to port next (Phase B):

Page	Source File	Key Feature
Odontogram	client/src/pages/OdontogramPage.tsx	Interactive tooth chart, condition coding
Perio Chart	client/src/pages/PerioChartPage.tsx	6-point probing, bleeding, recession
Voice AI	client/src/pages/VoiceAIPage.tsx	Real-time transcription, SOAP notes
Patient Timeline	client/src/pages/PatientTimelinePage.tsx	Complete patient history view
Clinical Dashboard	client/src/pages/ClinicalDashboardPage.tsx	Daily schedule, alerts
AI Letter Writer	client/src/pages/LetterWriterPage.tsx	Necessity + appeal letters
Insurance Tools	client/src/pages/InsurancePage.tsx	Eligibility, EOB parsing
Recall Manager	client/src/pages/RecallPage.tsx	Automated recall sequences

SECTION 08

Revenue Opportunities

Identified value creation from the unified platform

\$797K

Outbound Recall Pipeline

127 patients > 6mo, no surgery

+\$108K

Cross-coding (monthly)

30 cases x \$3,600 avg

+\$222K

AR Recovery (monthly)

31-60d bucket via CareStack sync

+\$47K

No-show Recovery (monthly)

SMS+call within 2h

DSO Licensing Model (Year 2):

Tier	Locations	Price/mo	Monthly ARR
Starter	1-5	\$299	\$1,495
Growth	6-25	\$499	\$7,485-12,475
Enterprise	26-100	\$399	\$10,374-39,900
Franchise	100+	\$299	\$29,900+

At 25 DSO locations (\$499/mo): \$120,000/month recurring. At 100 locations (\$399/mo): \$399,000/month. Platform cost is nearly fixed — each new location is ~95% gross margin.

Competitive Whitespace Analysis

- [NO] Trust AI (Isaac PracticeOS, \$299/mo, Feb 2026) — comms AI only, no clinical AI
- [NO] Planet DDS DentalOS (Feb 2026 early access) — scheduling AI, no 3D prosthetics
- [NO] Overjet (\$186M raised) — X-ray AI only, no RCM, no comms, no prosthetics
- [NO] Carestream — imaging only, no practice management
- [YES] Fusion Unified OS — Only platform with clinical AI + RCM AI + comms AI + DSO mgmt

SECTION 09

HIPAA Compliance Architecture

How PHI is protected across all three merged systems

HIPAA compliance is not an add-on — it is baked into the fusion_crm architecture at the schema and agent runtime level. The merge preserves and extends these protections.

PHI Gating

PhiService validates every PHI access. Agents never touch the database directly. No PHI path exists from frontend to fullarch-ai GPU service.

Audit Trail

audit.access_log is append-only. Every PHI access, every agent action, every proposal is logged with actor, timestamp, and action type.

Scan Hashing

STL scan files are identified by HMAC-SHA256(person_uid, FULLARCH_SALT). The GPU service sees only opaque hex strings — it cannot identify patients.

Governed Agents

All agent proposals require human review before production execution. No autonomous mutations. See GOVERNED_AGENT_LEARNING_STRATEGY.md.

Multi-tenant Isolation

tenant.tenant + tenant.location enforce row-level security. DSO location A cannot access DSO location B's PHI.

Encryption at Rest

tenant.integration_credential stores all API keys encrypted. FULLARCH_SALT never committed to git — generated per environment.

Do NOT Interfere

- [WARNING] Active Linear missions ENG-343 through ENG-350 in fusion_crm .agents/orchestration/
- [WARNING] These are parallel Claude Code + Codex missions running concurrently
- [WARNING] Merge execution code goes in packages/ and apps/ only — do NOT modify .agents/
- [WARNING] CareStack credentials: OAuth2 password grant, IDP at <https://id.carestack.com>

SECTION 10

Environment Setup

Required environment variables and dependency additions

New environment variables required:

Variable	Example / Source	Required By
FULLARCH_AI_BASE_URL	https://api.fullarch-ai.runpod.io	fullarch_ai/client.py
FULLARCH_AI_API_KEY	RunPod API key (from RunPod dashboard)	fullarch_ai/client.py
FULLARCH_SALT	python -c 'import secrets; print(secrets.token_hex(32))'	fullarch_ai/hasher.py
ANTHROPIC_API_KEY	Anthropic Console (console.anthropic.com)	ai_tools/coding.py, letters.py

Python dependencies (add to requirements.txt):

```
httpx>=0.27.0
anthropic>=0.25.0
arq>=0.25.0
pydantic>=2.0.0
```

Node dependencies (add to package.json):

```
@anthropic-ai/sdk@^0.24.0
drizzle-orm@^0.30.0
zod@^3.22.0
```

Worker registration (apps/worker/main.py):

```
from apps.worker.jobs.insight_engine import insight_engine_cron

JOBS = [
    ...existing_jobs...,
    insight_engine_cron, # Add this line
]
```

SECTION 11

Immediate Next Steps

Ordered action items for the development team

TODAY

- Generate FULLARCH_SALT: `python -c "import secrets; print(secrets.token_hex(32))"`
- Add all 4 new env vars to GCP Secret Manager and local `.env`
- Install new dependencies: `pip install httpx anthropic arq`
- Register `insight_engine_cron` in `apps/worker/main.py`

WEEK 1

- Register 4 FastAPI routes: `POST /api/tools/prosthetic-recommendation`, `/suggest-codes`, `/necessity-letter`, `/appeal-letter`
- Test fullarch-ai client: `python -c "from packages.integrations.fullarch_ai import FullArchAIClient; ..."`
- Update `App.tsx` to add routes for `revenue-command` and `prosthetic-ai` pages
- Verify `audit.access_log` entries are written for all new tool calls

WEEK 2-3

- Port odontogram, perio-chart, voice-ai pages (highest clinical value)
- Build patient-timeline composite view using CareStack appointment history
- Wire insight engine pattern library to agent prompt construction
- Set up hypothesis review UI in admin panel

WEEK 4+

- DSO admin portal: location management, aggregate analytics
- Tenant-scoped insight isolation
- Billing integration, usage metering
- SOC 2 pre-assessment, penetration test

This specification was generated by the Fusion Unified Dental OS self-improving agent system based on analysis of `fusion_crm`, `full-arch-crm`, and `fullarch-ai` codebases. All code files referenced exist in `/home/user/workspace/fusion_unified/`.

Sources:

1. Precedence Research — Dental PMS Market — <https://www.precedenceresearch.com/dental-practice-management-software-market>
2. Precedence Research — U.S. DSO Market — <https://www.precedenceresearch.com/us-dental-support-organizations-market>
3. Trust AI / Isaac PracticeOS launch — <https://trustdentistry.ai/press-news>
4. Planet DDS DentalOS announcement — <https://www.businesswire.com/news/home/20260226387949/en/Planet-DDS-Unveils-DentalOS-AI-Agents-A-New-AI-Workforce-for-Dental-Operations>